

ACTIVITY 8 — LEARNER WORKSHEET**Root Cause Analysis with 5 Whys, Fishbone, and Evidence**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	60 min	ANALYZE	A3

Name	
Date	

What you are doing

Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

What you will produce: A 5 Whys chain ending at an actionable cause, a Fishbone diagram across the five 5M categories, and an evidence-tested cause shortlist.

Tools and techniques: 5 Whys tool, Fishbone tool, 5M categories, brainstorming, multi-voting

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- defect-counts.csv — 6 rows (defect_category, count)
- root-cause-evidence.csv — 8 rows (evidence_id, observation, source, supports_cause)

templates/ — the worksheets you complete:

- cause-evidence-test.csv (candidate_cause, evidence_that_supports_it, evidence_that_contradicts_it, verdict)
- fishbone-causes.csv (category, possible_cause, supported_by_evidence)
- five-whys.csv (why_level, question, answer, evidence_needed, evidence_found)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Test every candidate cause against the Lab 7 data before shortlisting it.

Your steps

1. Write the problem statement precisely — the effect you are explaining.
2. Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause.

Tool: <https://alfredang.github.io/5whys/>

3. Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement.

Tool: <https://alfredang.github.io/fishbone/>

4. Brainstorm candidate causes into each category — no evaluation during generation.
5. Use multi-voting to shortlist the most likely causes as a team.
6. Test each shortlisted cause against your Lab 7 data — does the evidence support it?
7. State why the team must not jump straight to solutions.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

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Step 6 — Test each shortlisted cause against your Lab 7 data — does the evidence support it?

Step 7 — State why the team must not jump straight to solutions.

Answers to the discussion prompts

1. Did your 5 Whys end at something you can actually act on, or at 'people are careless'?

2. Which shortlisted cause does the Lab 7 data actually support?

3. Why must the team not jump straight to solutions here?

Reflect & discuss

Think of a recent fix at work — was it aimed at a cause or a symptom?

What evidence would have proved it either way?

Check your work

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.